

E-Commerce Customer Purchase Behavior Analysis

Project Description

This project focuses on analyzing customer purchasing behavior within an e-commerce platform using Power BI. The objective is to understand how customers interact with products, identify purchasing patterns, evaluate product performance, and uncover factors that drive revenue growth. Using data cleaning, modeling, DAX calculations, and interactive visualizations, the project aims to generate actionable insights that support data-driven business decision-making.

Dataset Information

Dataset Name: Brazilian E-Commerce Public Dataset by Olist

Source: Kaggle

Data Format: CSV Files

Dataset Description: The Olist dataset contains real-world e-commerce transaction data collected from a Brazilian online marketplace between 2016 and 2018. It includes information related to customers, orders, products, sellers, payments, reviews, and order deliveries.

Tables Used in the Project:

- Customers Dataset
- Orders Dataset
- Order Items Dataset
- Products Dataset
- Payments Dataset
- Sellers Dataset
- Product Category Translation Dataset

Key Attributes Available:

- Customer ID
- Order ID
- Product ID
- Seller ID
- Order Date
- Product Category
- Payment Type

- Order Value
 - Freight Charges
 - Delivery Status
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Objectives

1. Data Preparation and Transformation

Tasks:

- Import all relevant e-commerce dataset files into Power BI.
 - Remove duplicate records and handle missing values.
 - Verify and correct data types across all tables.
 - Rename, split, and merge columns where necessary.
 - Standardize data formats and ensure data consistency.
 - Document all cleaning and transformation steps performed in Power Query.
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2. Data Modeling and Relationship Building

Tasks:

- Identify common keys between Customers, Orders, Products, Sellers, and Payments tables.
 - Create relationships between tables using appropriate primary and foreign keys.
 - Develop a star-schema model for efficient analysis.
 - Validate data integrity and relationship accuracy.
 - Create calculated columns where additional business context is required.
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3. Customer Purchase Behavior Analysis

Tasks:

- Identify high-value and repeat customers.
- Analyze customer purchase frequency and buying patterns.
- Calculate Average Order Value (AOV).

- d. Examine customer contribution to overall revenue.
 - e. Analyze preferred payment methods and purchasing trends.
 - f. Study monthly and seasonal purchasing behavior.
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4. KPI Development and Business Insight Generation

Tasks:

- a. Create DAX measures and KPIs for business performance evaluation.
 - b. Calculate Total Revenue, Total Orders, Total Customers, Average Order Value, and Revenue per Customer.
 - c. Analyze product category performance and sales contribution.
 - d. Identify top-performing products and revenue-generating categories.
 - e. Utilize AI visuals such as Key Influencers and Decomposition Tree to discover hidden patterns.
 - f. Generate actionable business insights and recommendations.
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5. Interactive Dashboard Development and Storytelling

Tasks:

- a. Design a multi-page interactive Power BI dashboard.
 - b. Create KPI cards, bar charts, line charts, matrix visuals, and trend analyses.
 - c. Implement slicers, filters, drill-through functionality, and tooltips.
 - d. Incorporate Smart Narrative and AI-powered insights.
 - e. Ensure the dashboard follows a clear and engaging storytelling approach.
 - f. Present findings through a professional and visually appealing dashboard.
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Expected Outcomes

- 1. Identification of customer purchasing patterns and trends.
- 2. Recognition of top-performing products and revenue-generating categories.
- 3. Better understanding of customer retention and repeat purchase behavior.

4. Discovery of factors influencing sales and business growth.
5. Generation of actionable recommendations to improve customer engagement and revenue performance.
6. Development of a professional multi-page Power BI dashboard that supports strategic business decision-making.